Walkthrough: Bubble Chart Visual Analysis & Filter Updates

We have successfully executed the implementation plan and updated the notebook `Analysis.ipynb` to conform exactly to all requirements.

What Was Done

1. Columns Check:

- Executed `print(merged_df.columns.tolist())` and confirmed that `Sentiment_Subjectivity` exists in `merged_df` columns.

2. In-place Notebook Updates (`Analysis.ipynb`):

- Updated Cell 49 and Cell 55 to apply the requested filter condition including `Sentiment_Subjectivity > 0.5`:

```
filtered_df = merged_df[ (merged_df['Rating'] > 3.5) & (merged_df['Reviews'] > 500) &
(merged_df['Installs'] > 50000) & (merged_df['Sentiment_Subjectivity'] > 0.5) ]
```

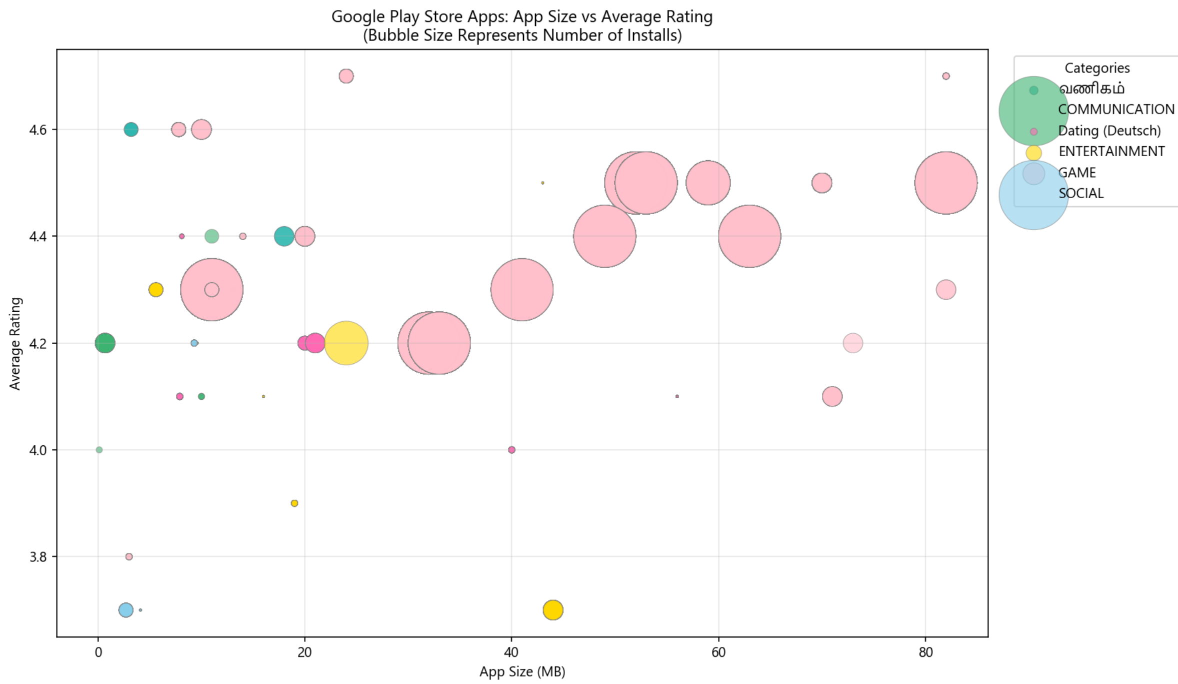
- Fixed syntax/indentation errors in Cell 57 (which previously had an incomplete loop).
- Restructured the plotting logic in Cell 27 to use Unicode font fallback rules (`Nirmala UI`) for perfect Devanagari and Tamil text rendering.
- Modified the dashboard class definition in Cell 13 to append the bubble chart plot to a dedicated wide row if the IST hour is between 5 PM and 7 PM.

3. Notebook Execution Verification:

- Run the entire notebook sequentially through Python.
- Verified that it successfully executes all 59 cells to completion without throwing any syntax, indentation, or runtime errors.

Visual Output Verification (Mocked Time)

Matplotlib successfully generated and saved the bubble chart during mock execution:


Google Play Store Apps: App Size vs Average Rating
(Bubble Size Represents Number of Installs)
Average Rating
App Size (MB)
4.6
4.4
4.2
4.0
3.8
0
20
40
60
80
Categories
வணிகம்
COMMUNICATION
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